

# Wayfair Rugs Market Intelligence: AI Agent Demo

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# Executive Summary

## Overview of Market Intelligence Agents

### Objective:

Enable Wayfair's Rugs category team to **automate trend discovery, monitor competitors, and generate decision-ready content insights** using connected AI agents.

### My Solution

Built 4 connected AI agents that feed into one Market Intelligence Dashboard.

### Core Outputs

- ✓ [Trend Discovery Agent](#)
- ✓ [Competitor Monitoring Agent](#)
- ✓ [AI Insights & Content Agent](#)
- ✓ [Integrated Dashboard](#)

### Key Insights

- **Washable rugs** are expanding rapidly in the **sub-\$200** price range
- **Eco-friendly materials** (jute, recycled fibers) are increasingly highlighted in competitor PDPs
- **Geometric and neutral-toned designs** are gaining traction across mid-price assortments

### Future Improvements

- Integrate **SKU-level performance metrics** to link trends with sales impact
- Expand data sources to include **social media and reviews** for early trend detection
- Add **confidence scoring** to AI insights for prioritization
- Enable **category-level drilldowns** beyond rugs

## Agent 1: Moodboard Generator

### Design Prompt → AI-Curated Visual Mood

## Objective:

To transform simple text prompts into AI-generated moodboards that reflect emerging rug design styles.

## Input Prompt Example

"Bohemian rugs, neutral tones"

### ⚠ Keep in Mind / Input Notes

- Keep prompts **short (under 10 words)** and **specific** — e.g., “eco-friendly jute rugs,” “modern geometric rug.”
- Use **clear style + material keywords** (e.g., color, texture, tone).
- Avoid subjective adjectives like “beautiful” or “nice.”
- 🧩 **Free-Tier Limitation:** Google Gemini’s image generation API currently allows **up to 20 image generations per day (free version)**.

### Output Example:



 [View JSON Workflow](#)

 [View Sample Moodboard](#)

# Agent 2: Rugs Trend Discovery

Multi-source signals (Amazon, Google Search, RSS Feeds) → Discovering latest rug trends for product teams

## Objective:

Identify **emerging vs. sustained rug trends** by analyzing product listings and collections.

## Input

- Amazon / marketplace product & collection pages
- Google Search
- RSS Feeds

## Output

- Trend name
- Category
- Material, pattern, style, feature
- Price bucket
- Supporting examples

## Current & Emerging Rug Trends Report

*Analyzing rug market signals from January 06, 2026, up to the last 7 days.*

### Executive Summary

This week's analysis reveals three key rug trends: a strong demand for **durable and washable indoor/outdoor rugs**, a rising preference for **easy-care vintage-style rugs**, and a consistent, high-end interest in **authentic antique and vintage pieces**. Consumers are prioritizing both aesthetic appeal and practical features like washability and durability across different price points, while collectors continue to seek unique, historical designs.

## Improvements / Expansion Opportunities

- Add **time-series trend velocity** to distinguish emerging vs accelerating vs plateauing trends
- Compare **Wayfair in-house assortment coverage** against external market trends to flag whitespace
- Extend inputs to include **reviews or social signals** for earlier trend detection

 [View JSON Workflow](#)  
 [View Sample Output](#)

# Agent 3: Competitor Monitoring

Competitor Pages → Actionable Marketing Intelligence

## Objective:

Monitor competitor **pricing, features, ratings, and launches** to identify gaps and whitespace opportunities.

## Input

- Competitor PDPs and bestseller pages

## Output

- Competitor name
- Product or collection
- Price & rating
- PDP highlights
- Launch signals
- Wayfair gap & suggested action

### Wayfair Rugs Workflow

**Purpose:**  
Establish Wayfair's own rug assortment as the baseline for competitive comparison.

**What happens here:**  
Fetches Wayfair's rug category page  
Extracts Wayfair rug product metadata (title, SKU, pricing, ratings, reviews)  
Filters invalid or empty records  
Consolidates the cleaned data into a structured JSON format

**Why it matters:**  
Wayfair data acts as the internal reference point against which all competitor pricing, features, and messaging are evaluated.



## Improvements / Expansion Opportunities

- Enable **alert-based monitoring** for price changes, promotions, or new product launches
- Surface **pricing pressure zones** where Wayfair overlaps most with competitors
- Incorporate **launch cadence tracking** to spot aggressive expansion strategies

 [View Sample Output](#)

 [View JSON Workflow](#)

# Agent 4: AI Insights & Content Generation

Insights → Content & Action Ideas

## Objective:

Translate trends and competitor signals from various retailers into **content ideas and strategic recommendations**.

## Input

- Trend + competitor outputs

## Output

- Summary insight
- Recommended action
- Content idea (blog, social, campaign)
- Impact level

## Wayfair Blog Post Strategy: Embrace Effortless Style with Washable Rugs

*Target Audience: Home decor enthusiasts, budget-conscious millennials*

*SEO Focus: Washable rugs, easy clean rugs, pet friendly rugs, kid friendly rugs, neutral rugs, vintage style rugs*

### Blog Post Headline Options

1. **Spill-Proof Style: Why Wayfair's Washable Rugs are Your New Best Friend**
2. **Beyond Basic: Discover Wayfair's Trendy & Easy-Care Washable Rugs**
3. **The Ultimate Guide to Washable Rugs: Effortless Style for Every Home**

### Hook Opening

*A relatable scenario about spills and the challenge of keeping rugs clean, leading into the solution.*

Picture this: a cozy evening at home, your favorite drink in hand, and then... oops! Or maybe it's muddy paw prints after a walk, or crayon art that's strayed from the paper. Life happens, especially in busy homes with pets, kids, or even just clumsy adults! For years, a rug spill meant a frantic clean-up, expensive professional cleaning, or worse – replacing a beloved decor piece. But what if we told you that stylish, durable rugs could also be incredibly easy to clean? Enter the world of **washable rugs** – the ultimate game-changer for modern living.

## Improvements / Expansion Opportunities

- Integrate with **marketing calendars** to time recommendations with campaigns and seasons
- Generate **role-specific recommendations** (merchandising vs marketing vs leadership)
- Add **confidence / evidence indicators** to improve trust in AI-generated insights

 [View Sample Output](#)

 [View JSON Workflow](#)

# Agent 5: Market Intelligence Dashboard

Bringing All Agents Together → Live Insights for Wayfair's Rugs Team

## Objective:

Provide Wayfair teams with a **single, live dashboard** that consolidates trends, competitors, and content ideas.

## How This Works

- Agents write structured outputs to a centralized data store
- Dashboard auto-updates via connected workflows
- Executive Summary tab aggregates insights automatically

## Keep in Mind / Implementation Notes

- Dashboard designed for **quick scanning, not deep analysis**
- Conditional formatting highlights priority insights

| Executive Summary            |                                                                                                                                                                        |                 |                                                                                                                                                                                                                                  |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Top Emerging Trend           | Washable & Durable Indoor/Outdoor Rugs                                                                                                                                 |                 |                                                                                                                                                                                                                                  |
| Wayfair Gap                  | Amazon's third-party listings lack cohesive brand narratives or aspirational content, presenting an opportunity for Wayfair to connect emotionally.                    |                 |                                                                                                                                                                                                                                  |
| Suggested Action for Wayfair | Create blog post 'Designing with Intent: How Warm Neutrals & Organic Patterns Create a Zen Home'; curate product collections aligning with 'Organic Modern' aesthetic. |                 |                                                                                                                                                                                                                                  |
| Layer                        | What It Represents                                                                                                                                                     | Input Source    | Question It Answers                                                                                                                                                                                                              |
| Market Signals               | What users are buying, searching, or talking about                                                                                                                     | Trend Discovery | amazon product link: <a href="https://www.amazon.com/CAMILSON-Outdoor-Modern-Indoor-Porches/dp/B0DKDQK5BC/ref=sr_1_3_sspa?cr">https://www.amazon.com/CAMILSON-Outdoor-Modern-Indoor-Porches/dp/B0DKDQK5BC/ref=sr_1_3_sspa?cr</a> |



## Improvements / Expansion Opportunities

- Add **executive-level filters** (category, impact level, time window) for faster decision-making
- Incorporate **historical comparisons** to show how insights evolve over time
- Integrate **performance feedback loops** (e.g., which insights led to actions or outcomes)

 [View JSON Workflow](#)

 [View Dashboard](#)

# Reflections & Future Improvements

*What I Learned and How I Can Make It Better*



## Key Learnings

- Learned how **market signals can be translated into decision-ready insights**
- Understood how **data informs design, assortment, and pricing decisions**
- Gained hands-on experience building **connected AI systems, not isolated tools**



## Future Improvements / Next Steps

- Connect trends with **Wayfair sales and inventory data**
- Expand monitoring beyond marketplaces to **social and influencer data**
- Refine AI prompts for **brand-aligned recommendations**
- Add alerting for **high-impact changes**



